

Finite Element Assessment of a Two-Link Robotic Arm Under Multiphysics Loading

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1. Objective

The objective of this study was to evaluate the structural integrity and dynamic performance of the two-link robotic arm developed in previous weeks. Five finite element analyses were performed using ANSYS Workbench:

- Transient Structural Analysis
- Explicit Dynamics Impact Analysis
- Fatigue Analysis
- Steady-State Thermal Analysis
- Random Vibration Analysis

The results were used to assess the suitability of the design under realistic operating conditions.

2. Model Description

The robotic arm consists of a fixed base connected to two links using revolute joints. Pin bodies were suppressed during analyses and the joints were represented using ANSYS revolute joint definitions. The analyses were performed using the same CAD model developed in the previous weeks.

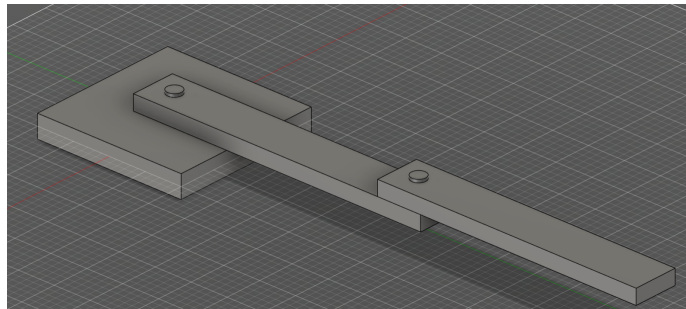


Figure 1: CAD Model

3. Transient Structural Analysis

Objective

To simulate the robotic arm sweeping from 0° to 90° in **0.5 s** under gravity and determine the dynamic stresses and joint reaction forces.

Boundary Conditions

- Fixed support applied to the base
- Revolute joints at shoulder and elbow
- Standard Earth gravity applied
- Base joint rotated from 0° to 90° over 0.5 s

Results

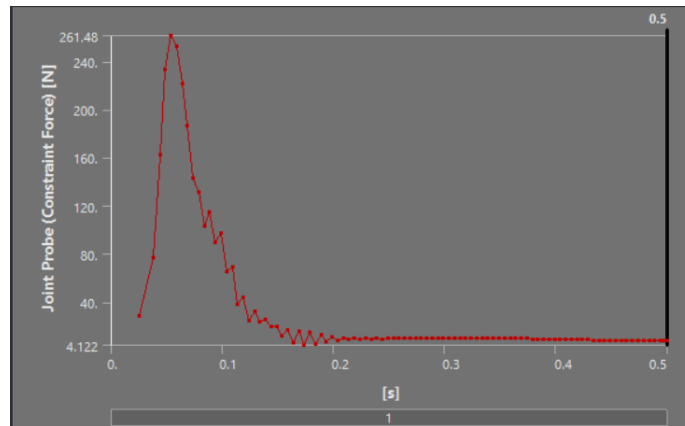


Figure 2: Joint Reaction Force

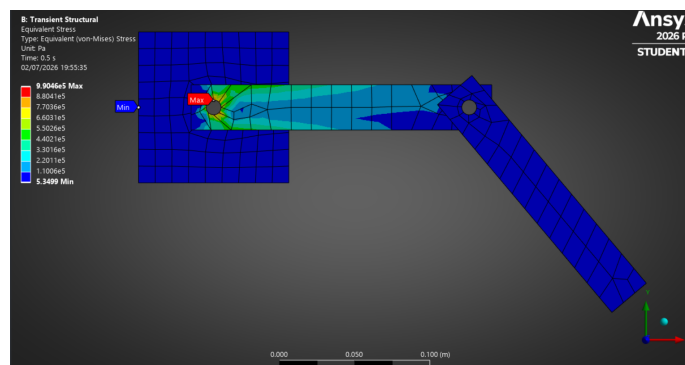


Figure 3: Peak Dynamic Stress

Observation

The arm completed the prescribed motion successfully without instability. The peak dynamic stress was slightly higher than the static stress obtained in Week 3 due to inertial loading during motion.

4. Explicit Dynamics

Objective

To evaluate impact behaviour when the end-effector strikes a rigid stop.

Boundary Conditions

- Fixed support applied to the base
- Second joint locked
- Rigid, fixed wall placed in front of end-effector
- Initial angular velocity = **4.48 rad/s** (End Effector velocity = 1.5 m/s)

Results

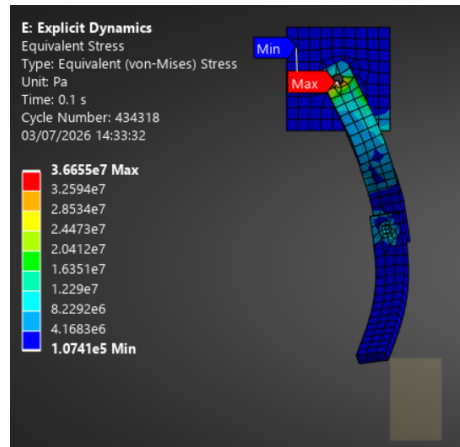


Figure 4: Peak Impact Stress

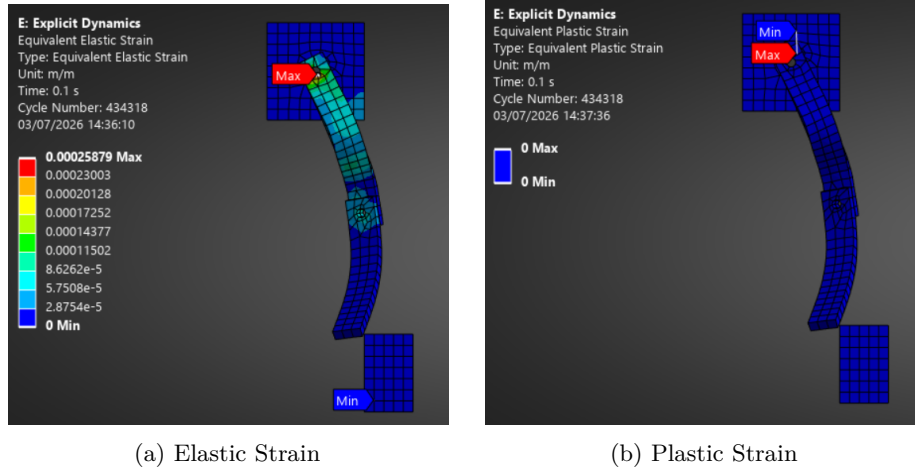


Figure 5: Impact Strain Analysis

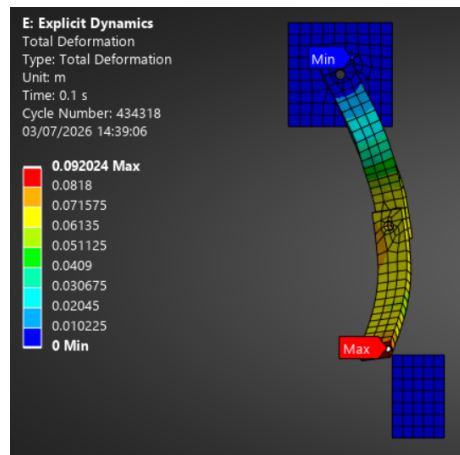


Figure 6: Total Impact Deformation

Observation

Peak stress of 36.7 MPa was observed at the moment of impact. The equivalent plastic strain remained zero, indicating that **no permanent deformation** occurred. Therefore, the impact response was entirely elastic, and the **material did not fail** under the specified loading conditions.

5. Fatigue Analysis

Objective

Estimate fatigue life under an alternating torque of $\pm 30 \text{ N} \cdot \text{m}$ at **1 Hz** using the Goodman correction.

Boundary Conditions

- Static Structural fatigue tool
- Goodman mean stress correction
- Alternating torque loading

Results

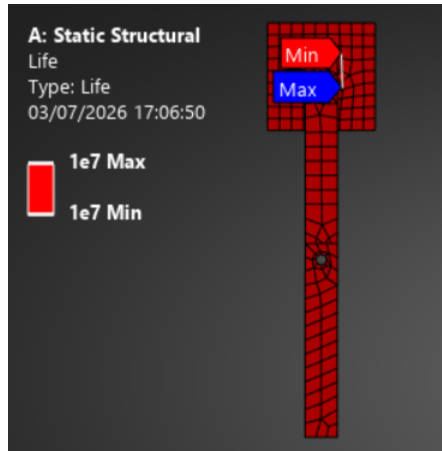


Figure 7: Predicted Fatigue Life

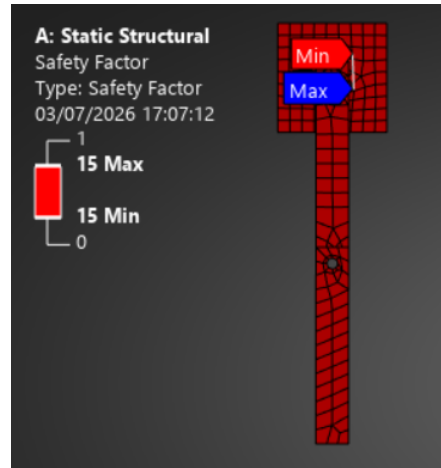


Figure 8: Factor of Safety Distribution

Observation

The fatigue analysis predicted a minimum life of 10^7 **cycles**, satisfying the required ten million cycle design life. The minimum fatigue **factor of safety was greater than 1**, indicating that the alternating stress remains below the material's allowable fatigue limit. Therefore, the shaft is considered safe for long-term cyclic operation under the specified loading conditions.

6. Thermal Analysis

Objective

Evaluate the temperature distribution caused by two 40 W motors and determine the resulting thermal stresses.

Boundary Conditions

- Internal heat generation equivalent to 40 W at the base and links
- Natural convection
- Ambient temperature = 22 °C

Results

Temperature

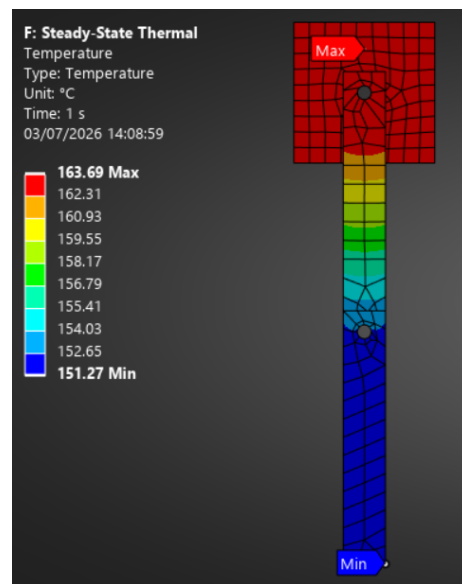


Figure 9: Temperature Distribution Map

Maximum Temperature = **163 °C**

Minimum Temperature = **151 °C**

Thermal Stress

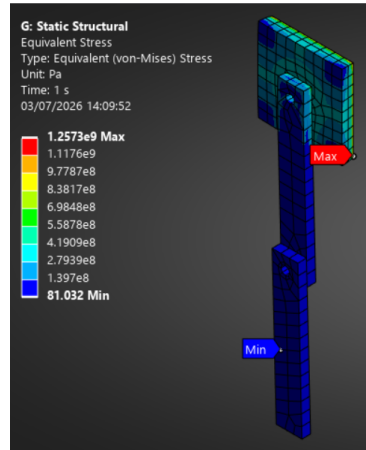


Figure 10: Thermal Equivalent Stress

Maximum Equivalent Stress = **1.25 GPa** (localized at the fixed base corners)

Thermal Deformation

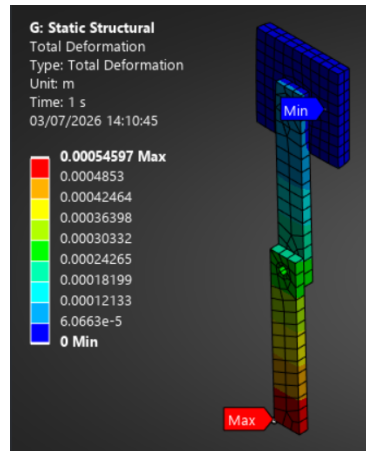


Figure 11: Total Thermal Deformation

Maximum Deformation $\approx$ **0.5 mm**

Observation

The maximum thermal stress occurred at the sharp corners of the fixed base due to stress concentration (numerical singularity). Despite the localized peak stress, the overall thermal deformation remained small (≈ 0.5 mm), indicating limited thermal expansion of the structure.

7. Random Vibration Analysis

Objective

Evaluate the vibration response under a flat acceleration PSD of **0.01 g²/Hz** between **20 Hz and 500 Hz**.

Modal Results

Mode	Frequency (Hz)
1	0
2	0
3	72.88
4	438.98
5	1233.2
6	1239.9

Modes 3 and 4 lie within the excitation frequency range and therefore dominate the random vibration response.

Random Vibration Results

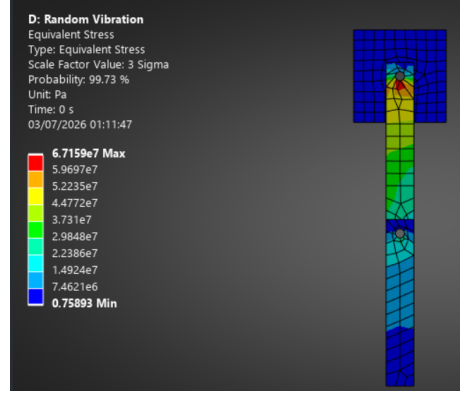


Figure 12: 3σ Equivalent Stress

- Maximum 3σ Equivalent Stress = **67 MPa**

Comparison with Yield Strength

Quantity	Value
Yield Strength	250 MPa
Maximum 3σ Stress	67 MPa

Since the maximum 3σ stress (67 MPa) is well below the material yield strength (250 MPa), the robotic arm is safe against yielding under the specified random vibration environment.

8. Conclusion

A comprehensive structural, dynamic, thermal and vibration assessment of the robotic arm was successfully completed. The transient analysis demonstrated stable operation during motion, while the explicit dynamics simulation confirmed elastic impact behaviour. Fatigue analysis predicted a life of at least 10^7 **cycles**, satisfying the design requirement. Thermal analysis showed moderate temperature rise with localized stress concentrations at constrained corners due to idealized boundary conditions. Random vibration analysis produced a maximum **3σ stress of 67 MPa**, well below the material yield strength.

Overall, the robotic arm meets the required structural performance criteria under all loading conditions considered.